

REHABILITATION, RECONSTRUCTION & RECOVERY

# Understanding India's Hazard Vulnerability: Earthquakes, Cyclones, and Floods

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India stands at a geographical crossroads where natural forces converge with devastating potential. While covering just two percent of the world's landmass, [approximately 85 percent of Indian land is vulnerable to one or more natural hazards](#), exposing over a billion people to recurring disasters. From the seismically active Himalayas to the cyclone-prone coastlines, understanding India's hazard vulnerability is essential for building resilient communities and effective disaster management systems.

## Table of Contents

1. India's vulnerability landscape

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2. Regional disaster risks across India

- a. The Himalayan belt: Earthquakes and landslides
- b. Coastal regions: Cyclones and storm surges
- c. River basins: Flooding challenges
- d. Drought-prone regions: Water scarcity zones

3. Impacts on communities and infrastructure

- a. Human toll and displacement
- b. Infrastructure and economic damage
- c. Agricultural and livelihood impacts

4. Building preparedness through institutional frameworks

- a. Legislative and institutional mechanisms
- b. Early warning systems and technological interventions
- c. Disaster response capabilities
- d. Community awareness and capacity building

5. The path forward: Integrated risk reduction

## India's vulnerability landscape

India's unique geographical position creates a complex hazard profile that makes it one of the world's most disaster-prone nations. Twenty-seven of its 29 states and seven union territories face exposure to recurrent natural hazards including earthquakes, cyclones, floods, droughts, and landslides. The numbers paint a sobering picture: 57 percent of India's landmass is vulnerable to earthquakes, 12 percent to floods, and 8 percent to cyclones, while 68 percent of cultivable land faces drought risk.

Climate change has compounded these vulnerabilities, increasing both the frequency and intensity of disasters. The country experiences over 50 significant natural disasters annually, affecting approximately 200 million people. The India Disaster Knowledge Network estimates that 50 million people are affected by droughts and 30 million by floods each year, creating immense pressure on communities, infrastructure, and the economy.

## Regional disaster risks across India

### The Himalayan belt: Earthquakes and landslides

The Himalayan region faces multi-hazard vulnerability due to its location along major fault lines. The entire belt falls under seismic zones IV and V, indicating potential for earthquakes exceeding magnitude 8.0. Over 59 percent of India's land area is under moderate to severe seismic hazard, with states like Jammu and Kashmir, Himachal Pradesh, Uttarakhand, and northeastern states at particular risk. The region is also highly susceptible to landslides, with approximately 15 percent of India's landmass prone to this hazard, and 80 percent concentrated in the Himalayan region.

Historical earthquakes demonstrate this vulnerability clearly. The 1905 Kangra earthquake claimed 19,800 lives, while the 2001 Bhuj earthquake in Gujarat resulted in 12,932 deaths and devastated 890 villages. The continuing northward movement of the Indian plate creates persistent stress on rocks, making them weak and susceptible to sudden release of energy.

### Coastal regions: Cyclones and storm surges

India's extensive coastline stretches over 7,500 kilometers, of which approximately 5,700 kilometers are prone to cyclones and storms from the Bay of Bengal and the Arabian Sea. The east coast faces particularly severe cyclone risk from May to June and October to November. States like Andhra Pradesh, Odisha, West Bengal, and Tamil Nadu along the Bay of Bengal experience frequent and intense cyclones. The west coast, including parts of Maharashtra and Gujarat facing the Arabian Sea, experiences less frequent but potentially more intense cyclonic storms.

Recent cyclones like Fani in Odisha (2019) and Amphan in West Bengal (2020) have caused widespread devastation, highlighting the region's vulnerability. The increasing sea surface temperatures due to climate change are intensifying the frequency and destructive power of tropical cyclones, making coastal preparedness more critical than ever.

### River basins: Flooding challenges

Flooding constitutes 46 percent of all disaster events in India and contributes the highest economic losses. Out of 329 million hectares of geographical area, more than 40 million hectares are flood-prone. The northern river systems, particularly the Ganges and Brahmaputra and their tributaries, cause recurring floods during monsoon season. States like Bihar, Uttar Pradesh, and Assam face annual flooding that disrupts millions of lives.

The 2005 Mumbai floods, which deposited 944 millimeters of rainfall in a single day, demonstrated urban vulnerability to extreme precipitation. The 2013 Uttarakhand disaster and 2018 Kerala floods further highlighted how unplanned urbanization, encroachment on floodplains, and inadequate drainage systems exacerbate flood vulnerability. On average, 7.5 million hectares are flooded annually, 1,600 lives are lost, and damage amounts to significant economic losses.

### Drought-prone regions: Water scarcity zones

While drought events occur less frequently than floods, they have widespread and devastating effects. A single drought event impacts an average of 75 million people, affecting far more individuals than floods. States like Rajasthan, Gujarat,

Maharashtra, Karnataka, and Telangana experience drought conditions every three to four years. These recurring droughts have profound socioeconomic implications, including agricultural distress, water scarcity, forced migration, and increased farmer suicides.

The 2015–2016 drought affected ten states and impacted an estimated 330 million people, including 37 million children under five years old. Heat waves accompanying droughts, particularly in regions like Vidarbha, Telangana, and Andhra Pradesh where temperatures exceed 45°C, create additional health hazards for vulnerable populations.

## Impacts on communities and infrastructure

### Human toll and displacement

Natural hazards create cascading impacts that extend far beyond immediate physical damage. [Between 2000 and 2019, India reported 79,732 deaths from natural disasters across 321 catastrophic events](#). On average, India loses approximately 3,600 lives annually to disasters, with earthquakes and floods causing the highest fatalities. Beyond mortality, disasters displace an average of 1.5 million people yearly, often leading to protracted recovery periods and secondary health crises.

Vulnerable groups bear disproportionate impacts. During the 2004 tsunami, female mortality in some coastal communities was three times higher than male mortality due to socio-cultural factors limiting their mobility. [In five major natural disasters from 2000 to 2016, some 17,671 children lost their lives](#), highlighting how disasters particularly affect the most vulnerable members of society.

### Infrastructure and economic damage

India's rapid urbanization and increasing population density in risk areas have amplified disaster impacts on infrastructure. Unscientific construction practices, multistoried apartments built without seismic considerations, and encroachment on floodplains create compound vulnerabilities. The 2001 Bhuj earthquake caused estimated economic damage of \$2.6 billion, while [India suffered nearly \\$80 billion in economic losses over 20 years from natural disasters](#).

Critical infrastructure including roads, bridges, power supply systems, communication networks, and water treatment facilities suffer extensive damage during disasters. Agricultural infrastructure also faces severe impacts, with crops washed away by floods, destroyed by cyclones, or withered by droughts. These losses ripple through the economy, affecting livelihoods, food security, and development progress.

### Agricultural and livelihood impacts

Agriculture, which employs a significant portion of India's population, remains highly vulnerable to natural disasters. Floods wash away or ruin crops, droughts cause widespread crop failure, and cyclones destroy standing crops and agricultural infrastructure. The economic impacts extend beyond immediate crop losses to include livestock deaths, soil degradation, and long-term productivity declines. Small and marginal farmers, who lack resources for recovery, face the most severe livelihood disruptions, often leading to distress migration and deepening poverty cycles.

## Building preparedness through institutional frameworks

### Legislative and institutional mechanisms

The Disaster Management Act of 2005 marked a paradigm shift in India's approach to disasters, moving from reactive relief to proactive preparedness. The Act established the [National Disaster Management Authority \(NDMA\)](#), headed by the Prime Minister, as the apex body for disaster management. State Disaster Management Authorities (SDMAs) and District Disaster Management Authorities (DDMAs) implement disaster management frameworks at regional and local levels.

The National Policy on Disaster Management (2009) further reinforced this shift, promoting a culture of prevention, mitigation, preparedness, and response. [The National Disaster Management Plan \(2019\) provides a dynamic and actionable blueprint for disaster risk reduction across India](#), aligning with international frameworks like the Sendai Framework for Disaster Risk Reduction.

## Early warning systems and technological interventions

India has made significant strides in developing early warning systems for various hazards. [Under the National Cyclone Risk Mitigation Project \(NCRMP\), early warning systems have been installed in coastal states](#), proving highly effective in reducing cyclone-related casualties. The India Meteorological Department (IMD) provides weather forecasts and cyclone warnings, while the Indian National Centre for Ocean Information Services (INCOIS) operates a Storm Surge Early Warning System.

The [SACHET National Disaster Alert Portal](#) serves as India's first and only portal to publish official warnings for all disasters from authorized sources. This Common Alerting Protocol (CAP)-based integrated alert system disseminates near real-time early warnings through multiple technologies using [geo-intelligence](#). Citizens receive disaster alerts via SMS, mobile apps, and browser notifications, enabling timely evacuation and preparedness actions.

## Disaster response capabilities

The National Disaster Response Force (NDRF), established in 2006, comprises 16 battalions drawn from [Central Armed Police Forces](#). These specialized teams conduct search, rescue, evacuation, and relief operations during natural calamities. The NDRF has been at the forefront of numerous national operations, including major rescue efforts during the 2008 Kosi floods, 2014 Jammu and Kashmir floods, and 2015 Nepal earthquake.

[NDMA's Disaster Management Exercise \(DMEx\) Guidelines mandate regular simulation-based exercises to test disaster plans, coordination, and readiness](#). Major drills like Exercise Suraksha Chakra, conducted across 55 locations in Delhi-NCR, simulate large-scale disasters to evaluate emergency preparedness and inter-agency coordination.

## Community awareness and capacity building

Effective disaster management requires active community participation and awareness. [Under NCRMP, 24,007 government officials have been trained through 925 capacity-building trainings, while 68,988 community representatives have been trained on disaster response skills](#) including first aid, search and rescue, and shelter management.

Local innovations demonstrate the power of community-based approaches. In flood-prone areas like Bihar and Uttar Pradesh, villagers raise their houses and gardens above flood levels using [earthen mounds](#). The city of Pune implemented a comprehensive flood management system including drainage mapping, stream widening, and tax incentives for households recycling wastewater and using rainwater harvesting.

## The path forward: Integrated risk reduction

[Hazard mapping](#) and risk assessment form the foundation of effective disaster preparedness. NDMA's Earthquake Risk Indexing (EDRI) project assesses earthquake risks in Indian cities, evaluating hazard, vulnerability, and exposure to guide mitigation efforts. Similar initiatives for flood mapping, cyclone vulnerability assessment, and drought monitoring help identify high-risk areas and prioritize interventions.

Integration of disaster risk reduction into development planning ensures that new infrastructure projects consider hazard vulnerabilities. Building codes for earthquake-resistant construction, land-use management to prevent encroachment on floodplains, and climate-resilient agricultural practices reduce future disaster risks while supporting sustainable development.

Despite progress, challenges remain. Poor communication systems in rural areas, gaps in last-mile connectivity for early warnings, inadequate public awareness, and limited resources for preparedness measures constrain disaster management effectiveness. Addressing these gaps requires sustained investment in technology, infrastructure, capacity building, and community engagement across all levels of governance.

**What do you think?** How can communities in disaster-prone areas better prepare themselves for natural hazards? What role should technology play in strengthening India's early warning systems and disaster response capabilities?

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### 1 Reconstruction and Rehabilitation as Means of Development

- a. Importance of Disaster Mitigation
- b. Cost-Benefit Analysis
- c. Relationship between Disasters and Development
- d. The Relief-Rehabilitation-Development Continuum
- e. Operationalizing Linking of Relief and Rehabilitation with Development
- f. Rebuilding Civil Society
- g. Rehabilitation as a Bridge between Relief and Development

### 2 Damage Assessment

- a. Sample Surveys
- b. Epidemiological Surveillance
- c. Nutrition Centred Health Assessment
- d. Remote Sensing and Aerial Photography

### 3 Role of Various Agencies in Disaster Management and Development

- a. Framework for Coordination at the Governmental Level
- b. Relevance of Community Participation
- c. Role of Non-Governmental Organizations
- d. Role of Other Agencies in Disaster Management

### 4 Information Management Structure

- a. Role of Information Dissemination in Disaster Management
- b. Need for an Effective Electronic Media
- c. Communication System for Information Management

### 5 Parameters of Vulnerability

- a. Concept of Vulnerability
- b. Parameters of Vulnerability
- c. Vulnerability Reduction Strategies
- d. Sustainable Livelihood Framework

### 6 Development of Physical and Economic Infrastructure

- a. Developing Physical and Economic Infrastructure
- b. Environmental Infrastructure Development
- c. Sustainable Community Development
- d. Disaster Preparedness in Asia

### 7 Creation of Long-term Job Opportunities and Livelihood Options

- a. Concept of Livelihood
- b. Case Studies on Livelihood Opportunities
- c. Livelihood Approach to Reconstruction
- d. Livelihood Options: Challenges and Limitations

### 8 Funding Arrangements for Reconstruction

- a. Reconstruction Requirements
- b. Funding Arrangements
- c. Fiscal Discipline
- d. Role of International Donor Agencies
- e. Mobilization of Community for Resource Generation

## 9 Nature of Damage to Houses and Infrastructure due to Disasters

- a. Hazard Vulnerability in India 🚩
- b. Earthquake Prone Areas in India
- c. Nature of Damage to Houses in Earthquakes
- d. Tropical Cyclones in India
- e. Damage to Housing during Cyclones
- f. Nature of Floods in India
- g. Damage to Housing and Infrastructure due to Floods

## 10 Disaster Resistant House Construction

- a. Guidelines for Disaster Resistant Construction
- b. Traditional Disaster Resistant Construction Techniques
- c. Stone and Brick Buildings
- d. Damage to Reinforced Concrete Cement Buildings
- e. Building Codes and Standards
- f. Recent Advances in Housing Technology
- g. Agencies involved in Disaster Resistant Construction

## 11 Role of Housing / Building Authorities

- a. Rehabilitation and Reconstruction in the Aftermath of Disasters
- b. Role of Various Agencies in Reconstruction
- c. Governmental Agencies
- d. Non-Governmental Agencies
- e. International Agencies

## 12 Education and Awareness

- a. Concepts of Education and Training
- b. Significance of Education, Training, and Awareness in Disaster Management
- c. Role of the Media
- d. Participation of Stakeholders
- e. People's Participation in Disaster Rehabilitation and Awareness

## 13 The Philosophy of Coping with Disasters

- a. The Philosophy of Coping with Disasters
- b. Disaster Recovery Planning
- c. Humanising Disaster Recovery Efforts

## 14 Dealing with Victims' Psychology

- a. Dealing with the Human Psyche in the Aftermath of Disasters
- b. Stress Management
- c. Countering Trauma through Counselling

## 15 Role of Information Dissemination

- a. Reaching out to the Community
- b. Media and Disaster Management
- c. Role of the Media in Disaster Management: Contemporary Context
- d. Role of Civil Society Organisations in Information Dissemination

## 16 Participative Rehabilitation Process- Some Case Studies

- a. Linking Disasters to Development: A Case of Community-led Disaster Management in Nepal
- b. Malpa Landslide
- c. Latur Earthquake
- d. Bhuj Earthquake

- e. [Livelihood and Employment Restoration Programme in Orissa](#)

#### **17 Role of Various Agencies in Recovery Measures**

- a. [Role of Rural and Urban Local Bodies](#)
- b. [Role of NGOs in the Recovery Process](#)
- c. [The Government-NGO Cooperation](#)
- d. [Role of Community-based Organisations](#)

#### **18 Monitoring and Evaluation of Rehabilitation Work**

- a. [Significance of Monitoring and Evaluation](#)
- b. [Guiding Principles of Monitoring and Evaluation](#)
- c. [The Evaluation Criteria](#)

#### **19 Constraints in Monitoring and Evaluation**

- a. [Reasons for Inadequate Monitoring and Evaluation](#)
- b. [Constraints in Monitoring and Evaluation](#)
- c. [Types of Data Collection](#)

#### **20 Long-term Recovery**

- a. [Incorporating Local Needs in the Rehabilitation Process](#)
- b. [Translating Local Needs into Action: Preparation of a Local Community Plan](#)
- c. [Joint Action Planning and Implementation](#)

#### **21 Long-term Counter Disaster Planning**

- a. [Long-term Planning: Approach and Direction](#)
- b. [Long-term Community-based Counter Disaster Planning](#)
- c. [Issues in Sustainability](#)
- d. [Integration of Policy Issues in Community-based Disaster Management](#)

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